VM Lifecycle

Create, Deploy, Manage, Protect

Complete guide to managing KubeVirt VMs from creation to day-2 operations using VirtCraft CLI and v9s web UI.

CLI + Web — Two Interfaces, One Platform

Create & Deploy

Multiple ways to create VMs — templates, profiles, YAML, or the web wizard.

CLI: One-Command Creation

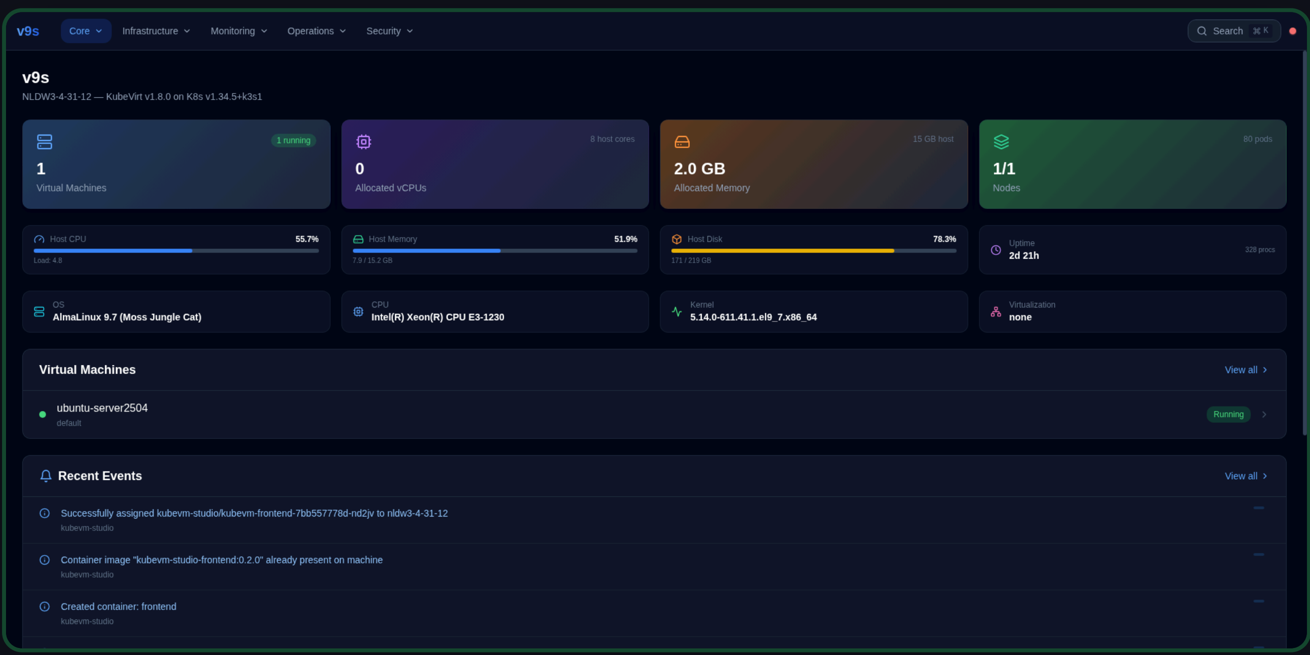
```
# From template + profile
$ virtcraft create --name db-01 --template rhel-9 --profile database

# With custom resources
$ virtcraft create --name web-01 --template ubuntu-24.04 \
    --cpus 4 --memory 8Gi --disk 50Gi --cloud-init user-data.yaml

# From YAML config file
$ virtcraft create -f vm-config.yaml

# Generate manifest without deploying
$ virtcraft generate --name test --template fedora-41 --profile dev > vm.yaml
```

Web: Dashboard VM View



VMs appear in the dashboard immediately after creation with real-time status updates.

Power Management

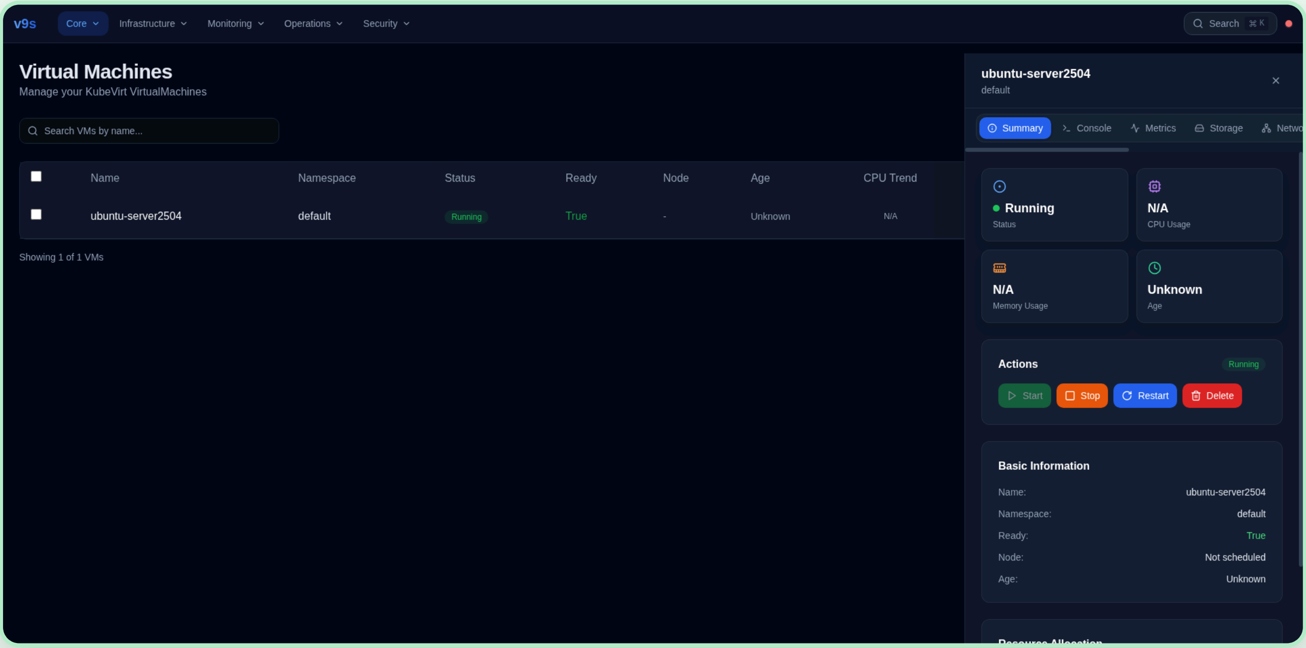
Start, stop, restart from CLI or web UI.

CLI Commands

```
$ virtcraft start web-01          # Start a VM
$ virtcraft stop web-01           # Graceful shutdown (ACPI)
$ virtcraft restart web-01        # Stop + start
$ virtcraft list                  # Check status

NAME     NAMESPACE  STATUS   RUNNING  CPU      MEMORY  IP          NODE
web-01   default    Running  Yes      4 cores  8Gi     10.42.0.15  node-1
db-01    default    Stopped  No       4 cores  16Gi    -           -
```

Web: Action Buttons

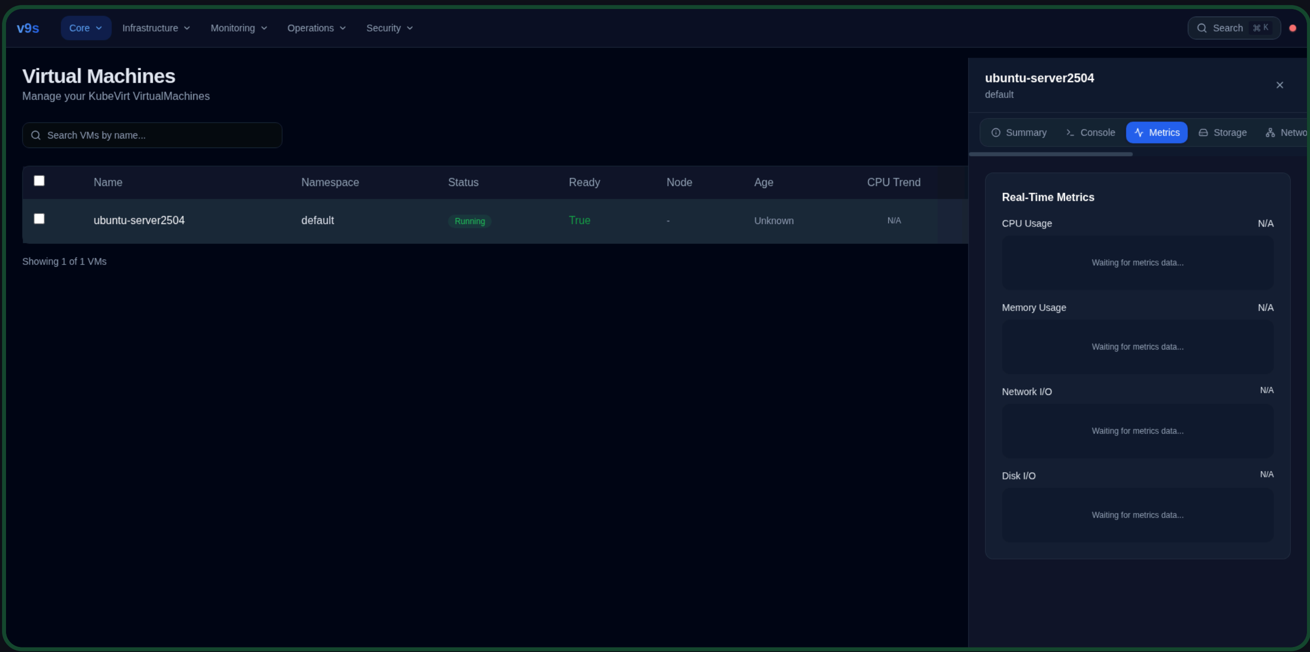


Start, Stop, Restart, Delete buttons with color coding. Status badge updates in real-time. Delete auto-closes the detail panel after success.

Monitor & Observe

Real-time metrics, events, and logs.

Live Metrics via WebSocket



CPU & Memory

Area charts with 60-point rolling history. Percentage display. Color-coded gradients.

Network & Disk I/O

Dual-line charts: RX/TX for network, Read/Write for disk. KB/s throughput tracking.

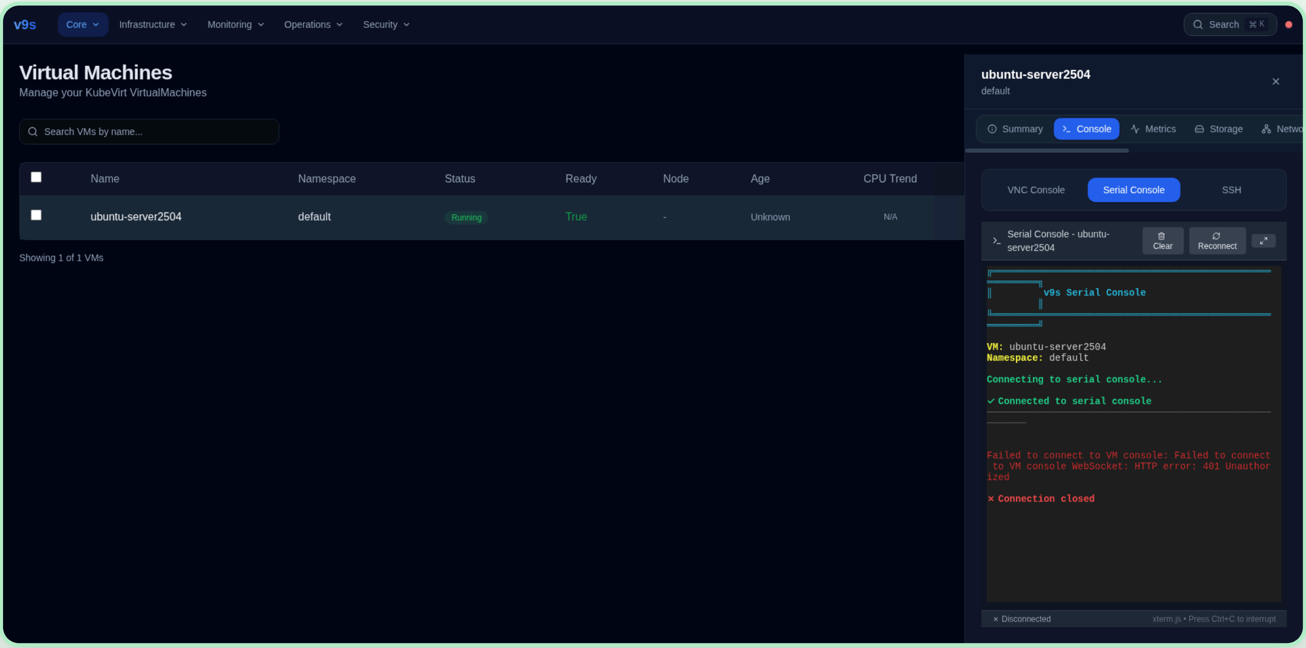
CLI Health Check

```
$ virtcraft health-check
  System Health Score: 87/100
  [OK]   CPU utilization: 45%
  [OK]   Memory utilization: 62%
  [WARN] Disk usage above 75% on db-01
  Recommendation: Expand disk on db-01 or archive old data

$ virtcraft resources          # Resource usage summary
$ virtcraft trends-analyze     # Usage trend analysis
```

Console Access

Three ways to access VMs directly.



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VNC Console

Full graphical desktop. noVNC via WebSocket. Sendkey: Ctrl+Alt+Del, Ctrl+Alt+F1/F2/F7. Fullscreen.

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Serial Console

xterm.js terminal. Boot troubleshooting. GRUB access. Kernel panic debugging. Headless VMs.

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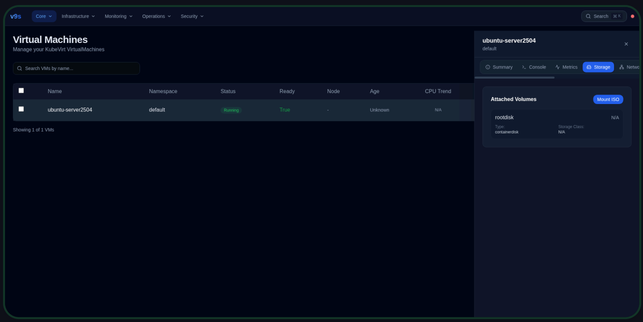
SSH Console

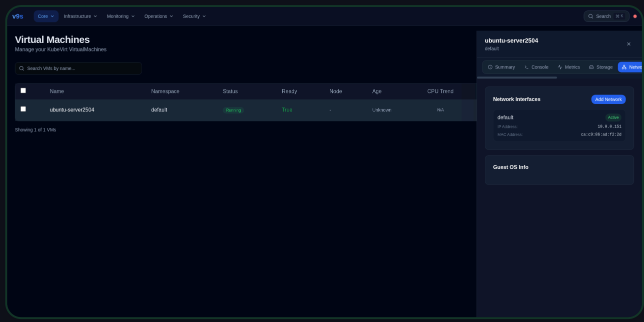
Web SSH via IP address. Configurable port. Fastest for CLI admin. Requires network connectivity.

Storage & Network

Disk management, ISO mount, and network interfaces.



Storage: volumes, ISO mount, storage classes

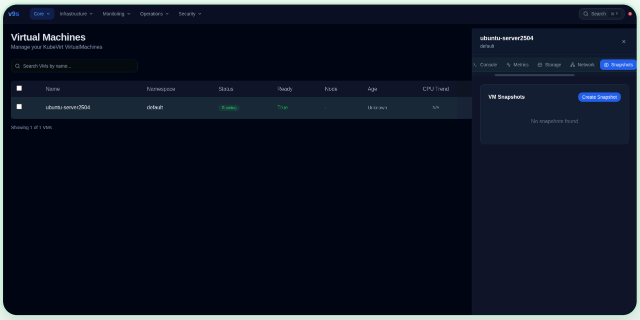


Network: interfaces, IP/MAC, add network

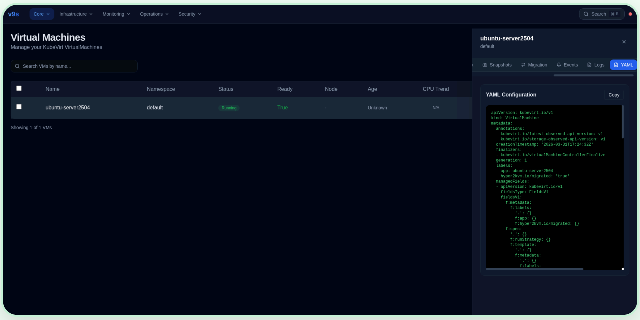
Operation	CLI	Web UI
View disks	virtcraft status vm-name	Storage tab
Mount ISO	YAML edit	Mount ISO button + modal
Add network	YAML edit	Add Network button + modal (pod/multus, bridge/masquerade/SR-IOV)
View interfaces	virtcraft get vm-name	Network tab (IP, MAC, state)
Guest OS info	N/A	Network tab (OS name, version, kernel)

Protect & Recover

Snapshots, cloning, and configuration backup.



Snapshots: create, list, delete with status



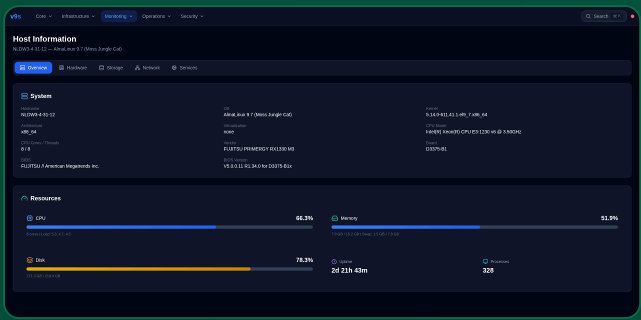
YAML: export, edit, copy configuration

CLI Snapshot & Clone

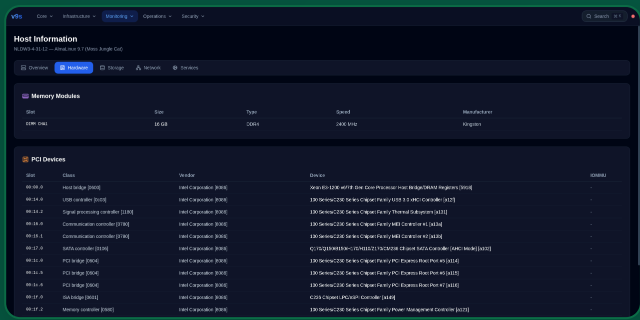
```
# Snapshots
$ virtcraft snapshot create web-01 --name pre-upgrade
$ virtcraft snapshot list web-01
$ virtcraft snapshot restore web-01 --name pre-upgrade

# Clone
$ virtcraft clone web-01 --name web-01-clone

# Export config
$ virtcraft export web-01 --format yaml > web-01-backup.yaml

# Config templates
$ virtcraft config-save --name "prod-web" --from web-01
$ virtcraft config-list
$ virtcraft config-load --name "prod-web" --as web-02
```

Host **Visibility**

Complete infrastructure monitoring from the web UI.



Overview: system info, CPU/memory/disk gauges



Hardware: RAM modules, PCI devices

Create → Deploy → Monitor → Console →

Protect → Scale

Complete VM lifecycle — CLI + Web

VirtCraft for power users. v9s for web management.
Same VMs, same cluster, two perfect interfaces.